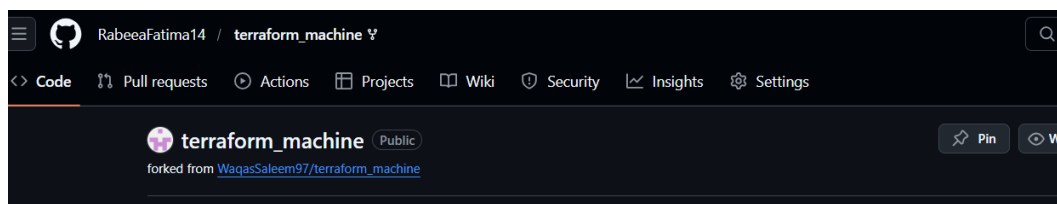


LAB NO 14
SUBMITTED BY:
RABEEA FATIMA
REGISTRATION NO: 2023-BSE-049
SECTION: B
SUBJECT: CLOUD COMPUTING

TASK 0:
Lab setup.



```
C:\Users\Hp>gh codespace list
NAME                DISPLAY NAME      REPOSITORY                BRANCH  STATE    CREATED AT
jubilant-doodle-7vj9q66944... jubilant doodle   RabeeaFatima14/Lab9      main*   Shutdown about 20 days
vigilant-meme-4jp7455799jg... vigilant meme      RabeeaFatima14/Lab10     main*   Shutdown about 19 days
supreme-waffle-97g9vrr94x5... supreme waffle    RabeeaFatima14/Assignment2 main*   Shutdown about 14 days
zany-rotary-phone-v6jv499v... zany rotary-phone RabeeaFatima14/Lab11     main*   Shutdown about 12 days
zany-space-yodel-pj5rv44rq... zany space yodel RabeeaFatima14/Lab12     main    Shutdown about 10 days
reimagined-space-goggles-w... reimagined space goggles RabeeaFatima14/Lab13    main    Shutdown about 10 days
solid-acorn-v6jv499v75q5cxwrj solid acorn       RabeeaFatima14/terrafor... main    Shutdown about 10 days

C:\Users\Hp>gh codespace ssh -c solid-acorn-v6jv499v75q5cxwrj
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ aws --version
```

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ aws --version
aws-cli/2.33.2 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
```

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible --version || echo "ansible not yet installed"
-bash: ansible: command not found
ansible not yet installed
```

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAXHUW46YMRMMH67S",
  "Account": "497458935321",
  "Arn": "arn:aws:iam::497458935321:user/Rabeea"
}
```

TASK 01:

Generate ssh key and Initial Terraform apply

1. Check SSH directory & generate SSH key pair if not already present

```
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ ls ~/.ssh
authorized_keys
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:vHBnSlee8mBZ/eqsttrpWkjbHzhVKyFQTgvh4Q276Ps codespace@codespaces-6d0678
The key's randomart image is:
+--[ED25519 256]--+
|      *oo         |
|    o X ..        |
|  + =0.. .        |
| .. .=...o..      |
|..S.X o...|
|.+ O B o..|
|.o o * o|
|. . o * .|
|..Eo=*o+|
+----[SHA256]-----+
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ ls -la ~/.ssh
total 24
drwxr-xr-x 2 codespace codespace 4096 Jan 17 09:00 .
drwxr-xr-x 1 codespace codespace 4096 Jan 17 08:59 ..
-rw-r--r-- 1 codespace codespace 163 Jan 17 08:48 authorized_keys
-rw----- 1 codespace codespace 419 Jan 17 09:00 id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan 17 09:00 id_ed25519.pub
```

2. Create terraform.tfvars in the repo root

```
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ cd /workspaces/terraform_machine
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ touch terraform.tfvars
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan 17 09:05 terraform.tfvars
```

```
Command Prompt - gh codespace ssh -c solid-acorn-v6jv499v75q5cxwj
GNU nano 7.2 terraform.tfvars *
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

3. Initialize terraform

```
@RabeeaFatima14 █ /workspaces/terraform_machine (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Finding latest version of hashicorp/http...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

4. Apply Terraform to create 2 EC2 instances

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be created
+ resource "aws_vpc" "myapp_vpc" {
  + arn              = (known after apply)
  + cidr_block       = "10.0.0.0/16"
```

5. Output

```
webserver_public_ips = [
  "3.28.219.36",
  "51.112.54.128",
]
```

TASK 02:

Static Ansible inventory with two EC2 instances

1. Install Ansible (core) using pipx

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ pipx install ansible-core
installed package ansible-core 2.20.1, installed using Python 3.12.1
These apps are now globally available
- ansible
- ansible-config
- ansible-console
- ansible-doc
- ansible-galaxy
- ansible-inventory
- ansible-playbook
- ansible-pull
- ansible-test
- ansible-vault
done!
```

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

2. Obtain the two public IPs of your EC2 instances

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.219.36",
  "51.112.54.128",
]
```

3. Create Ansible inventory file hosts

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ touch hosts
@RabeeaFatima14 /workspaces/terraform_machine (main) $ ls -la hosts
-rw-rw-rw- 1 codespace codespace 0 Jan 17 09:20 hosts
```

```

GNU nano 7.2                                     hosts *
3.28.219.36 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
51.112.54.128 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519

```

4. Test connectivity

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[ERROR]: Task failed: Failed to connect to the host via ssh: Host key verification failed.
Origin: <ad hoc 'ping' task>

{'action': 'ping', 'args': {}, 'timeout': 0, 'async_val': 0, 'poll': 15}

51.112.54.128 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}
3.28.219.36 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}

```

5. If it fails due to host key checking, add this to each line in hosts

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat hosts
3.28.219.36 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
51.112.54.128 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o StrictHostKeyChecking=no'

```

6. Retry

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '51.112.54.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See
https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.54.128 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See
https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.219.36 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

```

TASK 03:

Scale to three instances & group-based inventory

1. Update your Terraform module for webserver in main.tf to use count = 3

```

module "myapp-webserver" {
  source           = "./modules/webserver"
  env_prefix       = var.env_prefix
  instance_type    = var.instance_type
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id

  # Loop count
  count            = 3
  # Use count.index to differentiate instances
  instance_suffix = count.index
}

```

2. Apply Terraform to get 3 instances

```
Plan: 3 to add, 0 to change, 0 to destroy.
```

```
Changes to Outputs:
```

```
~ webserver_public_ips = [
  # (1 unchanged element hidden)
  "51.112.54.128",
  + (known after apply),
]
module.myapp-webserver[2].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[2].aws_security_group.web_sg: Creating...
module.myapp-webserver[2].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Creation complete after 3s [id=sg-0bd3c8291165b520f]
module.myapp-webserver[2].aws_instance.myapp-server: Creating...
module.myapp-webserver[2].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 12s [id=i-03b53c106117be7b7]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

3. Check outputs

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.219.36",
  "51.112.54.128",
  "158.252.95.169",
]
```

4. Rewrite your hosts file using group definitions

```
GNU nano 7.2
[ec2]
3.28.219.36
51.112.54.128

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
158.252.95.169

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'
```

5. Test group connectivity

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible ec2 -i hosts -m ping
[WARNING]: Host '51.112.54.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.54.128 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.219.36 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

6. Test single host by IP

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible 3.28.219.36 -i hosts -m ping
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.219.36 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

7. Test droplet group

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
[WARNING]: Host '158.252.95.169' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.95.169 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

8. Test all hosts

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '158.252.95.169' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.95.169 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.219.36 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '51.112.54.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.54.128 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

TASK 04:

Global ansible.cfg & first nginx playbook

1. Create global Ansible configuration

```
[default]
host_key_checking = False
interpreter_python = /usr/bin/python3
```

2. Remove ansible_ssh_common_args from hosts

```

GNU nano 7.2                                hosts *
[ec2]
3.28.219.36
51.112.54.128

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519

[droplet]
158.252.95.169

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed2551

```

3. Confirm connectivity

```

@RabeeaFatima14 [ /workspaces/terraform_machine (main) ] $ ansible all -i hosts -m ping
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.219.36 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '51.112.54.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.54.128 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.95.169' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.95.169 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

```

4. Create my-playbook.yaml

```

@RabeeaFatima14 [ /workspaces/terraform_machine (main) ] $ touch my-playbook.yaml
@RabeeaFatima14 [ /workspaces/terraform_machine (main) ] $ ls -la my-playbook.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 17 09:56 my-playbook.yaml

```

```

@RabeeaFatima14 [ /workspaces/terraform_machine (main) ] $ cat my-playbook.yaml
---
- name: Configure nginx web server
  hosts: ec2
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started

```


5. Run the playbook on [ec2] group

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.219.36]
[WARNING]: Host '51.112.54.128' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [51.112.54.128]

TASK [install nginx and update cache] *****
changed: [3.28.219.36]
changed: [51.112.54.128]

TASK [start nginx server] *****
changed: [3.28.219.36]
changed: [51.112.54.128]

PLAY RECAP *****
3.28.219.36      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
51.112.54.128   : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

6. Verify nginx default page on [ec2] servers by visiting http://<public-ip-ec2>



7. Change target to droplet

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat my-playbook.yaml
---
- name: Configure nginx web server
  hosts: droplet
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started
```

8. Re-run

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '158.252.95.169' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation
of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [158.252.95.169]

TASK [install nginx and update cache] *****
changed: [158.252.95.169]

TASK [start nginx server] *****
changed: [158.252.95.169]

PLAY RECAP *****
158.252.95.169 : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```


9. Verify nginx default page on droplet



TASK 05:

Single nginx target group & HTTPS prerequisites

1. Create project-level ansible.cfg in repo root

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ touch ansible.cfg
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan 17 10:09 ansible.cfg
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat ansible.cfg
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
```

2. Switch Terraform EC2 count back to 1

```
# Loop count
count          = 1
# Use count.index to differentiate instances
instance_suffix = count.index
```

3. Apply

```
module.myapp-webserver[1].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-1]
module.myapp-webserver[1].aws_security_group.web_sg: Destroying... [id=sg-09a32a095d1924c4e]
module.myapp-webserver[1].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-webserver[1].aws_security_group.web_sg: Destruction complete after 1s
Apply complete! Resources: 0 added, 0 changed, 6 destroyed.
```

4. Outputs

```
webserver_public_ips = [
  "3.28.219.36",
]
```

5. Adjust hosts

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat hosts
[nginx]
3.28.219.36

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

6. Update my-playbook.yaml

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat my-playbook.yaml
---
- name: Configure nginx web server
  hosts: nginx
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true
```

7. Run the playbook

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg
source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writa
ble-dir

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.219.36]

TASK [install nginx and update cache] *****
ok: [3.28.219.36]

TASK [install openssl] *****
ok: [3.28.219.36]

TASK [start nginx server] *****
changed: [3.28.219.36]

PLAY RECAP *****
3.28.219.36 : ok=4 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```

8. Verify nginx default page at http://<public-ip>



TASK 06:

Ansible-managed SSL certificates

1. Append this play after the nginx play in my-playbook.yaml

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ tail -n 50 my-playbook.yaml
hosts: nginx
become: true
tasks:
  - name: Create SSL private directory
    file:
      path: /etc/ssl/private
      state: directory
      mode: '0700'

  - name: Create SSL certs directory
    file:
      path: /etc/ssl/certs
      state: directory
      mode: '0755'

  - name: Get IMDSv2 token
    uri:
      url: "http://169.254.169.254/latest/api/token"
      method: PUT
      headers:
        X-aws-ec2-metadata-token-ttl-seconds: "3600"
      return_content: yes
    register: imdsv2_token

  - name: Get current public IP
```

2. Run

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg
source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writa
ble-dir

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.219.36]

TASK [install nginx and update cache] *****
ok: [3.28.219.36]

TASK [install openssl] *****
ok: [3.28.219.36]

TASK [start nginx server] *****
ok: [3.28.219.36]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [3.28.219.36]

TASK [Create SSL private directory] *****
changed: [3.28.219.36]

TASK [Create SSL certs directory] *****
changed: [3.28.219.36]

TASK [Get IMDSv2 token] *****
ok: [3.28.219.36]

TASK [Get current public IP] *****
ok: [3.28.219.36]

TASK [Show current public IP] *****
ok: [3.28.219.36] => {
  "msg": "Public IP: 3.28.219.36"
}

TASK [Generate self-signed SSL certificate] *****
changed: [3.28.219.36]

PLAY RECAP *****
3.28.219.36          : ok=11  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

3. SSH and verify:

TASK 07:

PHP front-end deployment with templates

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ mkdir -p files templates
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ touch files/index.php
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ touch templates/nginx.conf.j2
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ls -R
.:
README.md  files      locals.tf  modules    outputs.tf  terraform.tfstate  terraform.tfvars
ansible.cfg hosts      main.tf    my-playbook.yaml templates    terraform.tfstate.backup  variables.tf

./files:
index.php

./modules:
subnet  webserver

./modules/subnet:
main.tf  outputs.tf  variables.tf

./modules/webserver:
main.tf  outputs.tf  variables.tf

./templates:
nginx.conf.j2

```

2. Fill files/index.php with the following PHP metadata page

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat files/index.php
<?php
// Get hostname
$hostname = gethostname();

// Deployment date
$deployed_date = date("Y-m-d H:i:s");

// Metadata base URL
$metadata_base = "http://169.254.169.254/latest/";

// Function to get IMDSv2 token
function getImdsV2Token() {
    $ch = curl_init("http://169.254.169.254/latest/api/token");
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_CUSTOMREQUEST => "PUT",
        CURLOPT_HTTPHEADER => [
            "X-aws-ec2-metadata-token-ttl-seconds: 21600"
        ],
        CURLOPT_TIMEOUT => 2
    ]);
}

```

3. Fill templates/nginx.conf.j2 with the following Nginx configuration template

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat templates/nginx.conf.j2
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request"'
                   '$status $body_bytes_sent "$http_referer"'
                   '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile        on;
    tcp_nopush      on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include          /etc/nginx/mime.types;
}

```

4. Add this play to my-playbook.yaml

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ tail -n 40 my-playbook.yaml
creates: /etc/ssl/certs/selfsigned.crt

- name: Deploy Nginx website and configuration files
  hosts: nginx
  become: true
  tasks:
    - name: install php-fpm and php-curl
      yum:
        name:
          - php-fpm
          - php-curl
        state: present

    - name: Copy website files
      copy:
        src: files/index.php
        dest: /usr/share/nginx/html/index.php
        owner: nginx
        group: nginx
        mode: '0644'

    - name: Copy nginx.conf template
      template:
        src: templates/nginx.conf.j2
        dest: /etc/nginx/nginx.conf
```

5. Run the playbook

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg
source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writa
ble-dir

PLAY [Configure nginx web server] *****

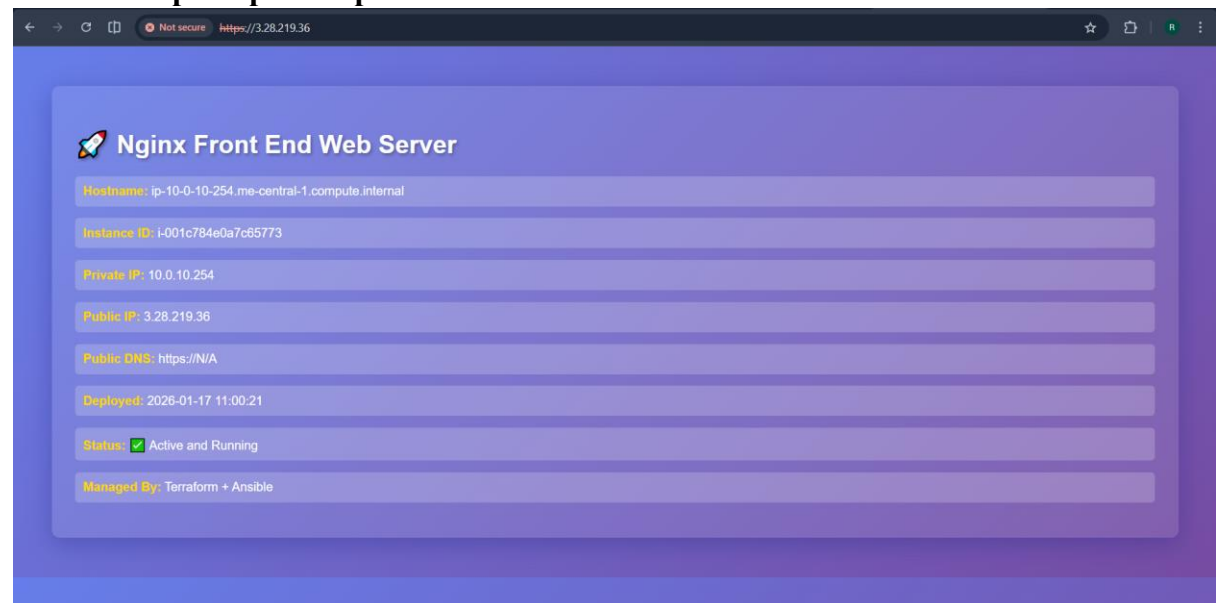
TASK [Gathering Facts] *****
[WARNING]: Host '3.28.219.36' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/
2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.219.36]

TASK [install nginx and update cache] *****
ok: [3.28.219.36]

TASK [install openssl] *****
ok: [3.28.219.36]

TASK [start nginx server] *****
```

6. Visit https://<public-ip>



TASK 08:

Docker & Docker Compose provisioning via Ansible

1. Destroy old infrastructure

```
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-040b9cb02fb856fbb]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0183d8b08fec4d97c]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0bcef52467d30130c]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
```

2. Recreate fresh infrastructure (1 instance)

```
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-070f1821af5504b37]
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-03d62101e0b7420c8]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-09d8dedb48bcd0e40]
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 13s [id=i-05e659bf84caf06ea]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
```

Outputs:

```
webserver_public_ips = [
  "3.28.74.5",
]
```

3. Update hosts file

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat hosts
[docker_servers]
3.28.74.5

[docker_servers:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

4. Replace my-playbook.yaml content

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat my-playbook.yaml
---
- name: Configure Docker
  hosts: all
  become: true
  tasks:
    - name: install docker and update cache
      yum:
        name: docker
        state: present
        update_cache: yes

- name: Install Docker Compose
  hosts: all
  become: true
  gather_facts: true
  tasks:
    - name: create docker cli-plugins directory
      file:
        path: /usr/local/lib/docker/cli-plugins
        state: directory
        mode: '0755'

    - name: install docker-compose
      get_url:
        url: https://github.com/docker/compose/releases/latest/download/docker-compose-linux-{{ lookup('pipe', 'uname -m') }}
```


5. Run the play

```
ubuntu@ip-10-0-10-158: /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir

PLAY [Configure Docker] *****
TASK [Gathering Facts] *****[WARNING]: Host '3.28.74.5' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.74.5]

TASK [Install docker and update cache] *****changed: [3.28.74.5]

PLAY [Install Docker Compose] *****
TASK [Gathering Facts] *****ok: [3.28.74.5]

TASK [create docker cli-plugins directory] *****changed: [3.28.74.5]

TASK [install docker-compose] *****changed: [3.28.74.5]

TASK [View architecture of the system] *****ok: [3.28.74.5] => {
  "msg": "System architecture of 3.28.74.5 is x86_64"
}

TASK [Alternate method to view architecture of the system] *****ok: [3.28.74.5] => {
  "msg": "System architecture of 3.28.74.5 is x86_64"
}

TASK [restart docker service] *****changed: [3.28.74.5]

PLAY RECAP *****3.28.74.5 : ok=0 changed=4 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```

6. Verify Docker

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ ssh ec2-user@3.28.74.5 -i ~/.ssh/id_ed25519

#_
~\ #### Amazon Linux 2023
~~ \#####\
~~ \###|
~~ \#/ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~
~~~_._\_/
~~~_/_/_/
~~~_/_/_/
~~~_/_/_/

Last login: Sat Jan 17 11:21:27 2026 from 20.192.21.53
[ec2-user@ip-10-0-10-158 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-10-0-10-158 ~]$ exit
logout
Connection to 3.28.74.5 closed.
```

TASK 09:

Gitea Docker stack via Ansible + Terraform security group update

1. Extend my-playbook.yaml

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ tail -n 30 my-playbook.yaml
- name: Adding user to docker group
  hosts: all
  become: true
  vars_files:
    - project-vars.yaml
  tasks:
    - name: add user to docker group
      user:
        name: "{{ normal_user }}"
        groups: docker
        append: yes

    - name: reconnect to apply group changes
      meta: reset_connection

    - name: verify docker access
      command: docker ps
      register: docker_ps
      changed_when: false

    - name: display docker ps output
      debug:
        var: docker_ps.stdout

    - name: fail if docker is not accessible
      fail:
        msg: "Docker is not accessible on this host"
        when: docker_ps.rc != 0
```

2. Create project-vars.yaml

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ touch project-vars.yaml
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ nano project-vars.yaml
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat project-vars.yaml
normal_user: ec2-user
docker_compose_file_location: /workspaces/terraform_machine
```

3. Add the deploy containers play

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ nano my-playbook.yaml
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ tail -n 40 my-playbook.yaml
- name: verify docker access
  command: docker ps
  register: docker_ps
  changed_when: false

- name: display docker ps output
  debug:
    var: docker_ps.stdout

- name: fail if docker is not accessible
  fail:
    msg: "Docker is not accessible on this host"
    when: docker_ps.rc != 0

- name: Deploy Docker Containers
  hosts: all
  become: true
  become_user: "{{ normal_user }}"
  vars_files:
    - project-vars.yaml
  tasks:
    - name: check if docker-compose file exists
      stat:
        path: /home/{{ normal_user }}/compose.yaml
      register: compose_file

    - name: copy docker-compose file
      copy:
        src: "{{ docker_compose_file_location }}/compose.yaml"
        dest: /home/{{ normal_user }}/compose.yaml
        mode: '0644'
      when: not compose_file.stat.exists

    - name: deploy containers using docker-compose
      command: docker compose up -d
      args:
        chdir: /home/{{ normal_user }}
      register: compose_result
      changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"
```

4. Create compose.yaml in repo root

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ touch compose.yaml
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ nano compose.yaml
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat compose.yaml
services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    extra_hosts:
      - "www.jenkins.com:host-gateway"
    networks:
      - webnet
  db:
    image: postgres:alpine
    container_name: gitea_db
    environment:
      - POSTGRES_USER=gitea
      - POSTGRES_PASSWORD=gitea
      - POSTGRES_DB=gitea
    restart: always
    volumes:
      - gitea_postgres:/var/lib/postgresql/data
    expose:
      - 5432
    networks:
      - webnet
volumes:
  gitea_postgres:
    name: gitea_postgres
  gitea:
    name: gitea
networks:
  webnet:
    name: webnet

```

5. Run playbook and visit the public-ip on your browser but webpage will not be shown due to permission issue

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir

PLAY [Configure Docker] *****

TASK [Gathering Facts] *****
[WARNING]: 'ansible' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.20.74.5]

TASK [Install docker and update cache] *****
ok: [3.20.74.5]

PLAY [Install Docker Compose] *****

TASK [Gathering Facts] *****
ok: [3.20.74.5]

TASK [Create docker-aliasing directory] *****

```

6. Update security group in modules/webserver/main.tf to add port 3000

```

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ nano modules/webserver/main.tf
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ grep -A 50 "resource \"aws_security_group\" \"web_sg\"" modules/web
server/main.tf
resource "aws_security_group" "web_sg" {
  name           = "${var.env_prefix}-web-sg-${var.instance_suffix}"
  description    = "Security group for web server allowing HTTP, HTTPS and SSH"
  vpc_id         = var.vpc_id

  ingress {
    description = "SSH from my IP"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    description = "HTTP"
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTPS"

```

7. Apply

```

    }
    name           = "dev-web-sg-0"
    tags           = {
      "Name" = "dev-default-sg"
    }
    # (9 unchanged attributes hidden)
  }
}

Plan: 0 to add, 1 to change, 0 to destroy.
module.myapp-webserver[0].aws_security_group.web_sg: Modifying... [id=sg-03d62101e0b7420c8]
module.myapp-webserver[0].aws_security_group.web_sg: Modifications complete after 1s [id=sg-03d62101e0b7420c8]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:
webserver_public_ips = [
  "3.28.74.5",
]

```

8. Access Gitea

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *

Host *

Username *

Password *

Database Name *

SSL *

Schema

Leave blank for database default ("public").

General Settings

Site Title *

You can enter your company name here.

Repository Root Path *

TASK 10:

Automating Ansible with Terraform (null_resource)

1. Add null_resource to main.tf

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ nano main.tf
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ tail -n 15 main.tf
    triggers = {
      webserver_public_ips_for_aws = join(",", [for i in module.myapp-webserver : i.public_ip])
    }

    depends_on = [module.myapp-webserver]

    provisioner "local-exec" {
      command = <<-EOT
        ansible-playbook -i ${self.triggers.webserver_public_ips_for_aws}, \
          --private-key "${var.private_key}" --user ec2-user \
            my-playbook.yaml
      EOT
    }
  }
}
```

2. Destroy and recreate infrastructure

```
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 1m21s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-09d8dedb48bcd0e40]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-03d62101e0b7420c8]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0c849e49c131fe7d5]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.

null_resource.configure_server (local-exec): TASK [deploy containers using docker-compose] *****
*
null_resource.configure_server: Still creating... [01m11s elapsed]
null_resource.configure_server: Still creating... [01m21s elapsed]
null_resource.configure_server (local-exec): ok: [3.28.46.176]

null_resource.configure_server (local-exec): PLAY RECAP *****
*
null_resource.configure_server (local-exec): 3.28.46.176          : ok=16   changed=6    unreachable=0    failed=0
skipped=1   rescued=0   ignored=0

null_resource.configure_server: Creation complete after 1m25s [id=2436820048538523024]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

Outputs:

webserver_public_ips = [
  "3.28.46.176",
]
```

3. If Ansible fails due to readiness, add a wait play at the top of my-playbook.yaml

```
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ nano my-playbook.yaml
@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ head -n 12 my-playbook.yaml
---
- name: Wait for some time to ensure system readiness
  hosts: all
  tasks:
    - name: Wait 300 seconds for port 22 to become open and contain "OpenSSH"
      wait_for:
        port: 22
        host: "{{ inventory_hostname }}"
        delay: 10
        timeout: 300
      delegate_to: localhost
```

4. Destroy and apply again

```
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0c8342bad7f8e93df, 00m40s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0c63730fb20c80f56, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0c8342bad7f8e93df, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 58s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0c63730fb20c80f56, 01m00s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 1m1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0a69a2cf79f2afa60]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-00a6cff38d8a63113]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 0s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0d1c150f532d73b6c]
aws_vpc.myapp_vpc: Destruction complete after 0s

Destroy complete! Resources: 8 destroyed.

null_resource.configure_server (local-exec): TASK [deploy containers using docker-compose] *****
*
null_resource.configure_server: Still creating... [01m20s elapsed]
null_resource.configure_server: Still creating... [01m30s elapsed]
null_resource.configure_server (local-exec): ok: [158.252.33.37]

null_resource.configure_server (local-exec): PLAY RECAP *****
*
null_resource.configure_server (local-exec): 158.252.33.37          : ok=17   changed=6   unreachable=0   failed=0
skipped=1   rescued=0   ignored=0

null_resource.configure_server: Creation complete after 1m34s [id=6295771713262946666]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

Outputs:

webserver_public_ips = [
  "158.252.33.37",
]
```

5. Verify Gitea / Nginx application is reachable at the appropriate URL

The screenshot shows the 'Initial Configuration' page for Gitea. It is divided into two main sections: 'Database Settings' and 'General Settings'. In the 'Database Settings' section, the 'Database Type' is set to 'PostgreSQL', 'Host' is 'db:5432', 'Username' is 'gitea', 'Password' is masked with '****', 'Database Name' is 'gitea', 'SSL' is set to 'Disable', and 'Schema' is left blank. A note states: 'Leave blank for database default ("public").'. In the 'General Settings' section, the 'Site Title' is 'Gitea: Git with a cup of tea' and the 'Repository Root Path' is '/data/git/repositories'. A note for the site title says: 'You can enter your company name here.'

TASK 11:

Dynamic inventory with aws_ec2 plugin

1. Update ansible.cfg

```

@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ nano ansible.cfg
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ cat ansible.cfg
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
deprecation_warnings = False

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519

```

2. Create inventory_aws_ec2.yaml

```

@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ touch inventory_aws_ec2.yaml
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 17 13:15 inventory_aws_ec2.yaml
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ nano inventory_aws_ec2.yaml
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ cat inventory_aws_ec2.yaml
---
plugin: aws_ec2
regions:
  - me-central-1

```

3. Ensure Terraform code includes both dev and prod webserver in main.tf

```

@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ nano main.tf
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ grep -A 15 "myapp-webserver" main.tf
module "myapp-webserver" {
  source           = "../modules/webserver"
  env_prefix       = var.env_prefix
  instance_type    = var.instance_type
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id

  # Loop count
  count            = 1
  # Use count.index to differentiate instances
  instance_suffix = count.index
}

--
module "myapp-webserver-prod" {
  source           = "../modules/webserver"
  env_prefix       = "prod"
  instance_type    = "t3.nano"
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id

  # Loop count
  count            = 1

```

4. Add outputs in outputs.tf

```

@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ nano outputs.tf
@RabeeaFatima14 ❏ /workspaces/terraform_machine (main) $ cat outputs.tf
output "webserver_public_ips" {
  value = module.myapp-webserver[*].public_ip
}

output "prod-webserver_public_ips" {
  value = module.myapp-webserver-prod[*].public_ip
}

```


5. Rebuild infra

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-webserver-prod in modules/webserver
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/null v3.2.4
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Creating...
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creating...
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Creation complete after 0s [id=prod-serverkey-0]
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-03ac4b78ea1c41756]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Creating...
module.myapp-webserver-prod[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Creation complete after 13s [id=i-00a373a3d75f38ffe]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

prod-webserver_public_ips = [
  "40.172.113.145",
]
webserver_public_ips = [
  "158.252.33.37",
]
```

6. Install boto3 and botocore

```
@RabeeaFatima14 /workspaces/terraform_machine (main) $ $(which python) -m pip install boto3 botocore
Collecting boto3
  Downloading boto3-1.42.30-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.30-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages (from boto3) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages (from botocore) (2.5.0)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-dateutil<3.0.0,>=2.1->boto3) (1.17.0)
Downloading boto3-1.42.30-py3-none-any.whl (140 kB)
Downloading botocore-1.42.30-py3-none-any.whl (14.6 MB)
 14.6/14.6 MB 30.8 MB/s 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.30 botocore-1.42.30 jmespath-1.0.1 s3transfer-0.16.0

@RabeeaFatima14 /workspaces/terraform_machine (main) $ $(which python) -c "import boto3, botocore; print(boto3.__version__)"
1.42.30
```

7. Check inventory graph

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils.text' is deprecated. This feature will be removed from an
ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils.text' is deprecated. This feature will be removed from
ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This feature will be removed from ansible-core ve
rsion 2.23.
@all:
  |--@ungrouped:
  |--@aws_ec2:
    |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
    |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com
```

TASK 12:

Filtering EC2 instances by tags & instance type

1. Modify inventory_aws_ec2.yaml to add tag-based grouping

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ nano inventory_aws_ec2.yaml
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat inventory_aws_ec2.yaml
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"

@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils.text' is deprecated. This feature will be removed from an
ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils.text' is deprecated. This feature will be removed from
ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This feature will be removed from ansible-core ve
rsion 2.23.
@all:
  |--@ungrouped:
  |--@aws_ec2:
    |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
    |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com
    |--@tag_Name_prod_ec2_instance_0:
      |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
    |--@tag_Name_dev_ec2_instance_0:
      |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com
```

2. Extend to group by instance type as well

```
@RabeeaFatima14 @ /workspaces/terraform_machine (main) $ cat inventory_aws_ec2.yaml
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"

  - key: instance_type
    prefix: instance_type
    separator: "_"
```

```

@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
  |   |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_ec2_instance_0:
  |   |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
  |--@instance_type_t3_nano:
  |   |--ec2-40-172-113-145.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_ec2_instance_0:
  |   |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com
  |--@instance_type_t3_micro:
  |   |--ec2-158-252-33-37.me-central-1.compute.amazonaws.com

```

3. Extend to group by instance type as well

```

@RabeeaFatima14 /workspaces/terraform_machine (main) $ head -n 20 my-playbook.yaml
---
- name: Configure nginx web server
  hosts: all
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started

```

4. Extend to group by instance type as well

```

TASK [install php-fpm and php-curl] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Copy website files] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]
changed: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-158-252-33-37.me-central-1.compute.amazonaws.com : ok=18  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
ec2-40-172-113-145.me-central-1.compute.amazonaws.com : ok=18  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0

```

5. Run only on dev instances

```

TASK [Gathering Facts] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [install php-fpm and php-curl] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Copy website files] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-158-252-33-37.me-central-1.compute.amazonaws.com : ok=18  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0

```

6. Run only on t3.micro instances

```

TASK [Gathering Facts] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [install php-fpm and php-curl] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Copy website files] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-158-252-33-37.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-158-252-33-37.me-central-1.compute.amazonaws.com : ok=18  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0

```

7. Run only on t3.nano instances

```

TASK [Copy website files] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-40-172-113-145.me-central-1.compute.amazonaws.com : ok=18  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
0 ignored=0

```

8. Update ansible.cfg to use inventory by default

```

@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat ansible.cfg
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
deprecation_warnings = False
inventory = ./inventory_aws_ec2.yaml

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519

```

9. Now you can simply run

```

@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ ansible-playbook -l instance_type_t3_nano my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-40-172-113-145.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****

```

CLEANUP

```

module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0938ddb2dd6dd5ff4, 01m00s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 1m0s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0403ec9e9dfdfc99]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Destruction complete after 1m0s
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Destroying... [id=prod-serverkey-0]
module.myapp-webserver-prod[0].aws_security_group.web_sg: Destroying... [id=sg-03ac4b78ea1c41756]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-06b7ed1d62b61ed63]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
module.myapp-webserver-prod[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0914b1d193e54f91a]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 11 destroyed.

```

```

@RabeeaFatima14 [ ] /workspaces/terraform_machine (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 105,
  "lineage": "7e77521c-26aa-43fb-495a-67993fda3dc7",
  "outputs": {},
  "resources": [],
  "check_results": null
}

```